

Can an LLM Follow the Evidence in a Graph?

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Quantitative Results Comparison

This section compares Method A (Neighborhood baseline) and Method B (RoG-inspired method) across eight fixed test questions from the RoG-WebQSP dataset. A result is counted as *strictly supported* only when every predicted answer is paired with a cited evidence path that starts from one of the reported linked entities and ends exactly at that answer.

Method A generated answers for “Bella Abzug” and “Madrid.” However, only the Madrid result satisfies the strict endpoint-binding rule: the Bella Abzug output includes additional descriptive strings and evidence paths whose endpoints do not exactly match the generated answers. Method B generated answers for six questions, and all six answer records satisfy the strict answer–path binding rule.

Table 1: Quantitative Results Comparison

Metric	Method A	Method B
Link Hit@1	87.5% (7 / 8)	87.5% (7 / 8)
Link Hit@3	87.5% (7 / 8)	87.5% (7 / 8)
Answer Records Generated	2 / 8	6 / 8
Strictly Supported Answer Records	1 / 8	6 / 8
Avg. Unique Triples Shown to LLM	50.0	17.00
Maximum Triples Shown to LLM	50	50

Failure Analysis

0.1 Entity Linking Mistake (Method A - Baseline)

- **Question:** ”who was the president after jfk died”
- **Analysis:** The question uses the common acronym ”jfk”, but the target node in the graph is formally labeled ”John F. Kennedy”. A purely character-based string matching algorithm (like ‘difflib.SequenceMatcher’) cannot recognize that these two strings refer to the identical semantic entity, as they share almost no character overlap. Without an external alias dictionary or a semantic vector representation, the linker completely misses the ground-truth entity.

Both the words ”president” and ”jfk” appear as exact substrings within the original question, triggering the ”exact match” condition and receiving a base score of 1.0. To break ties, the linker’s logic adds a tiny score bonus based on string length (favoring longer, ostensibly more specific entities). Because ”President” (9 characters) is longer than ”JFK” (3 characters), it receives a higher final score (1.009 vs 1.003) and incorrectly claims the top rank. The pure

string-based linker fails to distinguish between a generic noun/title ("President") and the actual core entity ("jfk").

0.2 Incomplete Answer Set Despite Correct Grounding (Method B)

- **Target Entity:** "Mozambique"
- **Analysis:** For the query "where is the Mozambique located," Method B returned "Africa" and cited the grounded evidence path:

Mozambique $\xrightarrow{\text{location.location.containedby}}$ Africa.

This answer is structurally supported: the cited path starts from the linked entity Mozambique and ends exactly at the predicted answer Africa. However, the dataset reference answer set contains both "Africa" and "Southern Sun." Therefore, Method B received only partial credit under exact answer-set evaluation because it omitted one valid reference answer.

- **Interpretation:** This is not an unsupported hallucination. The grounding and endpoint-validation constraints worked as intended: Method B did not output an answer without a matching path. Instead, the failure is one of answer completeness: among the available grounded endpoints, the final answering step selected only one valid answer. This distinction shows that evidence faithfulness and complete answer-set recall are separate evaluation dimensions.

Edge-Removal Test: Faithful Reasoning

To test whether Method B's final answer depends on its retrieved graph evidence, I conducted a controlled edge-removal intervention on a correctly answered multi-step question. The linker candidates and the successful relation plan from the original run were retained for the modified-graph run; Method B was not allowed to re-plan after the edge was removed.

- **Target Question:** "what was Liam Neeson's character in Star Wars"
- **Original Grounded Evidence Path:**

Liam Neeson $\xrightarrow{\text{film.actor.film}}$ m.0k3qyf $\xrightarrow{\text{film.performance.character}}$ Qui-Gon Jinn.

On the original graph, Method B returned "Qui-Gon Jinn" with this two-hop evidence path.

- **Removed Bridge Triple:**

[m.0k3qyf, film.performance.character, Qui-Gon Jinn]

The original graph contained 2,778 triples; the modified graph contained 2,777 triples after removing this single edge.

- **Method A Behavior:** Method A abstained on both the original and modified graphs. Therefore, its result is inconclusive for this intervention: it does not demonstrate a causal change in response to the removed edge.
- **Method B Behavior:** On the modified graph, the retained two-hop relation plan could no longer be fully grounded because its second edge was absent. Retrieval returned no complete evidence path, and Method B consequently returned an empty answer and the status **abstain**.
- **Interpretation:** In this controlled case, Method B changed from a correct, evidence-supported answer to abstention exactly when the cited supporting edge was removed. This behavior is

consistent with evidence-faithful reasoning: the system did not retain the original answer once a complete grounded path was unavailable. The experiment does not prove universal faithfulness or rule out every possible alternative route in other graphs, but it provides direct evidence that this answer was constrained by its retrieved graph path rather than emitted without support.

Paper Note: Reasoning on Graphs (RoG)

RoG makes LLM-based knowledge-graph question answering more evidence-grounded by separating planning, retrieval, and answering. Instead of reasoning directly over an unstructured sub-graph, the LLM first proposes a relation sequence; graph traversal then retrieves exact paths that match the plan, and the final answer is selected from those paths.

In this assignment, Method B uses the entity linker’s actual Top-3 candidates, retrieves executable relation paths, and exposes at most 50 unique triples to the answering LLM. Each answer is retained only if a cited path starts from a linked entity and ends exactly at that answer; otherwise, the system abstains. This design reduces unsupported generation, although it does not by itself guarantee complete answer sets.

Conclusion

The experiment shows that Method A’s fixed-size two-hop neighborhood retrieval introduces substantial local graph noise and depends entirely on its Top-1 entity-linking decision. In contrast, Method B uses the entity linker’s real Top-3 candidates, relation planning, and graph-constrained retrieval to focus the answering prompt on a smaller set of relevant evidence paths. Under the enforced budget, Method B never exposes more than 50 unique triples per question, averages 17.00 triples, and produces six answer records that satisfy the strict answer–path endpoint rule.

The Liam Neeson edge-removal test provides a controlled example of evidence-faithful behavior: once the bridge edge supporting “Qui-Gon Jinn” was removed, Method B could not retrieve a complete path and abstained. This result is evidence consistent with faithful reasoning for that intervention, rather than a universal proof of faithfulness.

Finally, the results distinguish evidence support from answer completeness. Method B’s Mozambique result was grounded but incomplete under exact answer-set evaluation, while the Lois Griffin and New Jersey cases returned empty answers and empty evidence paths with an **abstain** status. A robust KGQA system should retain strict path-based validation while improving plan selection and answer-set coverage.